



Taller de Docker- IIS

Entregables:

1. Motores DB docker
2. Bases de datos del proyecto en docker
3. Backend y Frontend IIS
4. Backend y Frontend en Docker apuntando a la base de datos Docker

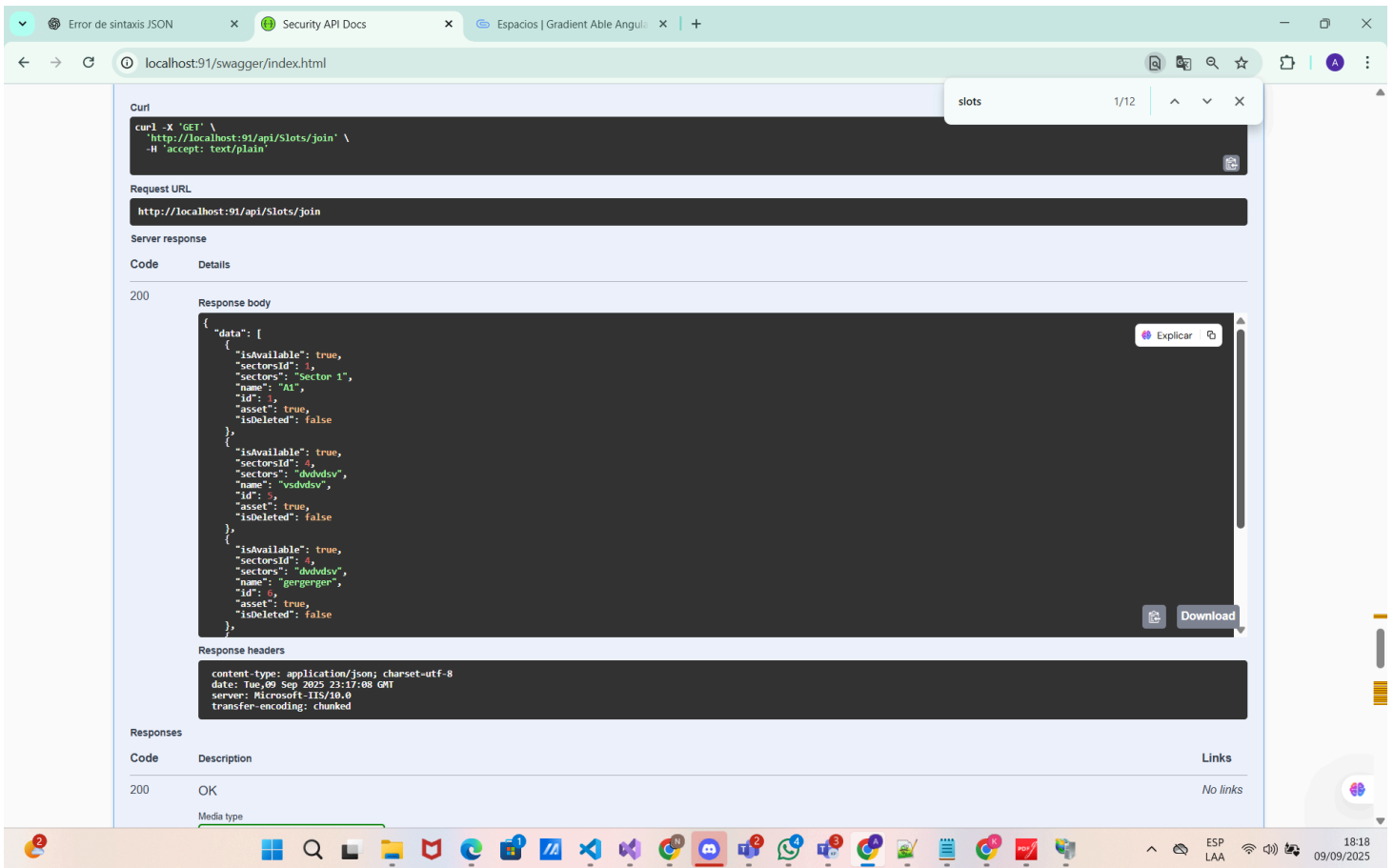
1. Backend y Frontend en IIS

En la imagen del Administrador de Internet Information Services (IIS) se observa que los sitios del proyecto fueron creados y están Iniciados con sus respectivos puertos y rutas físicas.

- Backend (AnprVision) desplegado correctamente en HTTP :91
- Frontend (AnprVisionFront) desplegado correctamente en HTTP :92

Nombre	Id.	Estado	Enlace	Ruta de acceso
AnprVision	3	Iniciado (h...)	*:91 (http)	C:\inetpub\wwwroot\backIIS
AnprVisionFront	4	Iniciado (h...)	*:92 (http)	C:\inetpub\wwwroot\front
Default Web Site	1	Iniciado (h...)	*:80 (http)	%SystemDrive%\inetpub\wwwroot
sded	2	Iniciado (h...)	*:90 (http)	C:\inetpub\wwwroot\back

Evidencias del back y front con sus respectivos puertos



The screenshot shows a web browser window with the address bar displaying `localhost:91/swagger/index.html`. The browser has several tabs open, including "Error de sintaxis JSON", "Security API Docs", and "Espacios | Gradient Able Angul...".

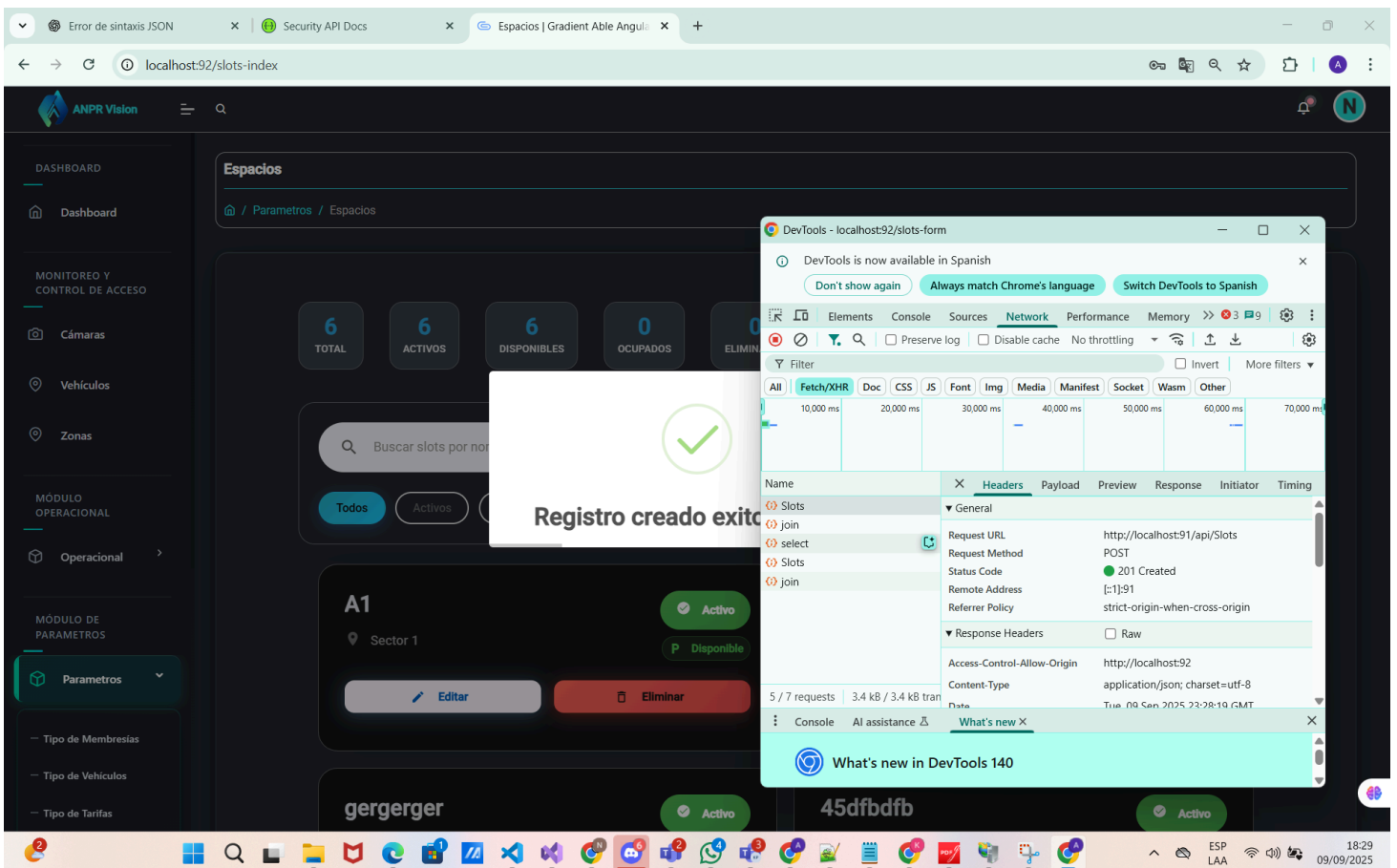
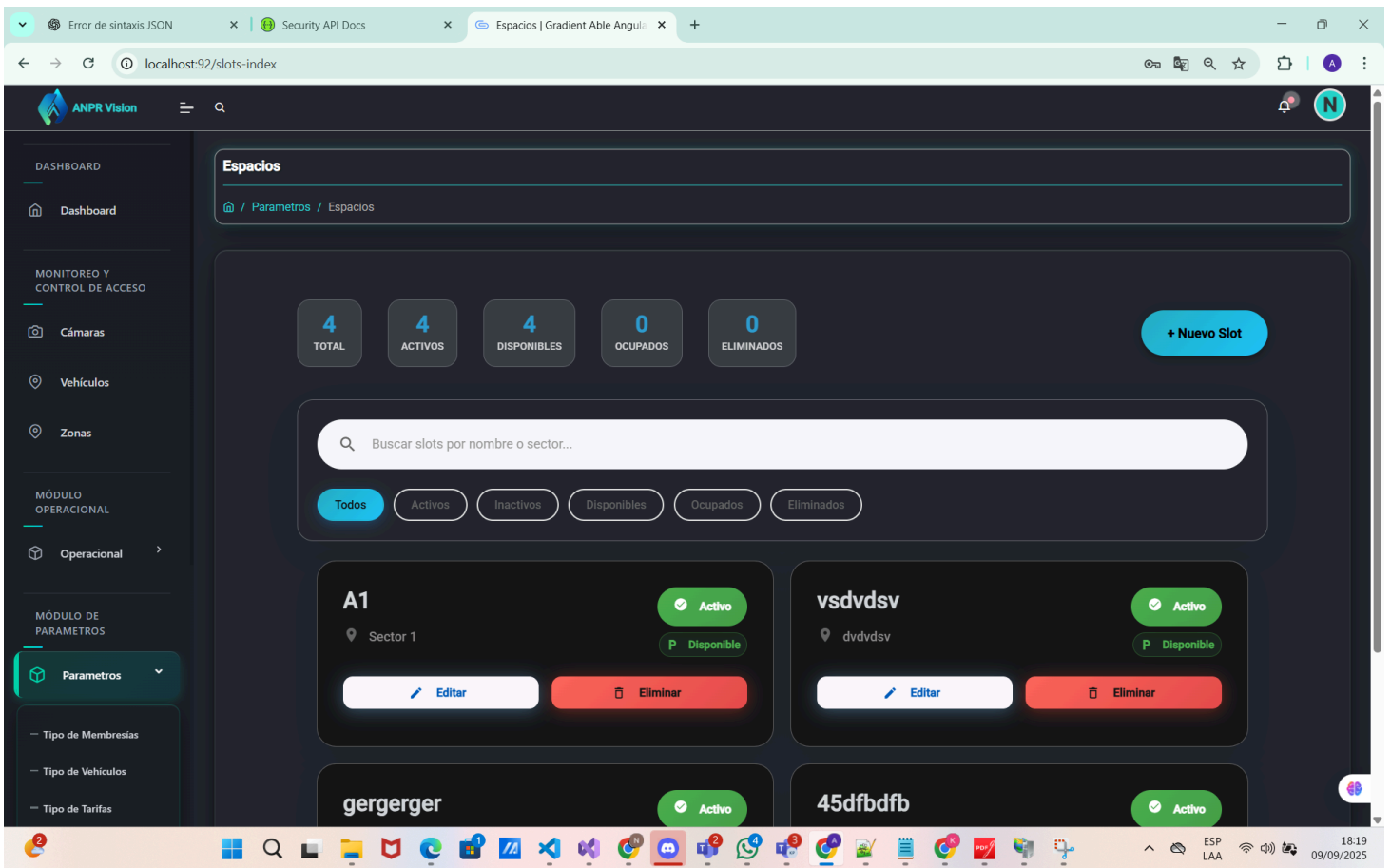
The main content area displays the Swagger API interface for the `slots` endpoint. The interface shows the following details:

- Request URL:** `http://localhost:91/api/Slots/join`
- Server response:** 200
- Response body:** A JSON array containing three objects, each representing a slot. The objects have the following structure:

```
{  "data": [    {      "isAvailable": true,      "sectorId": 1,      "sectors": "sector 1",      "name": "A1",      "id": 1,      "asset": true,      "isDeleted": false    },    {      "isAvailable": true,      "sectorId": 4,      "sectors": "dvdvdsv",      "name": "ysdvdsv",      "id": 5,      "asset": true,      "isDeleted": false    },    {      "isAvailable": true,      "sectorId": 4,      "sectors": "dvdvdsv",      "name": "gergerger",      "id": 6,      "asset": true,      "isDeleted": false    }  ]}
```
- Response headers:**

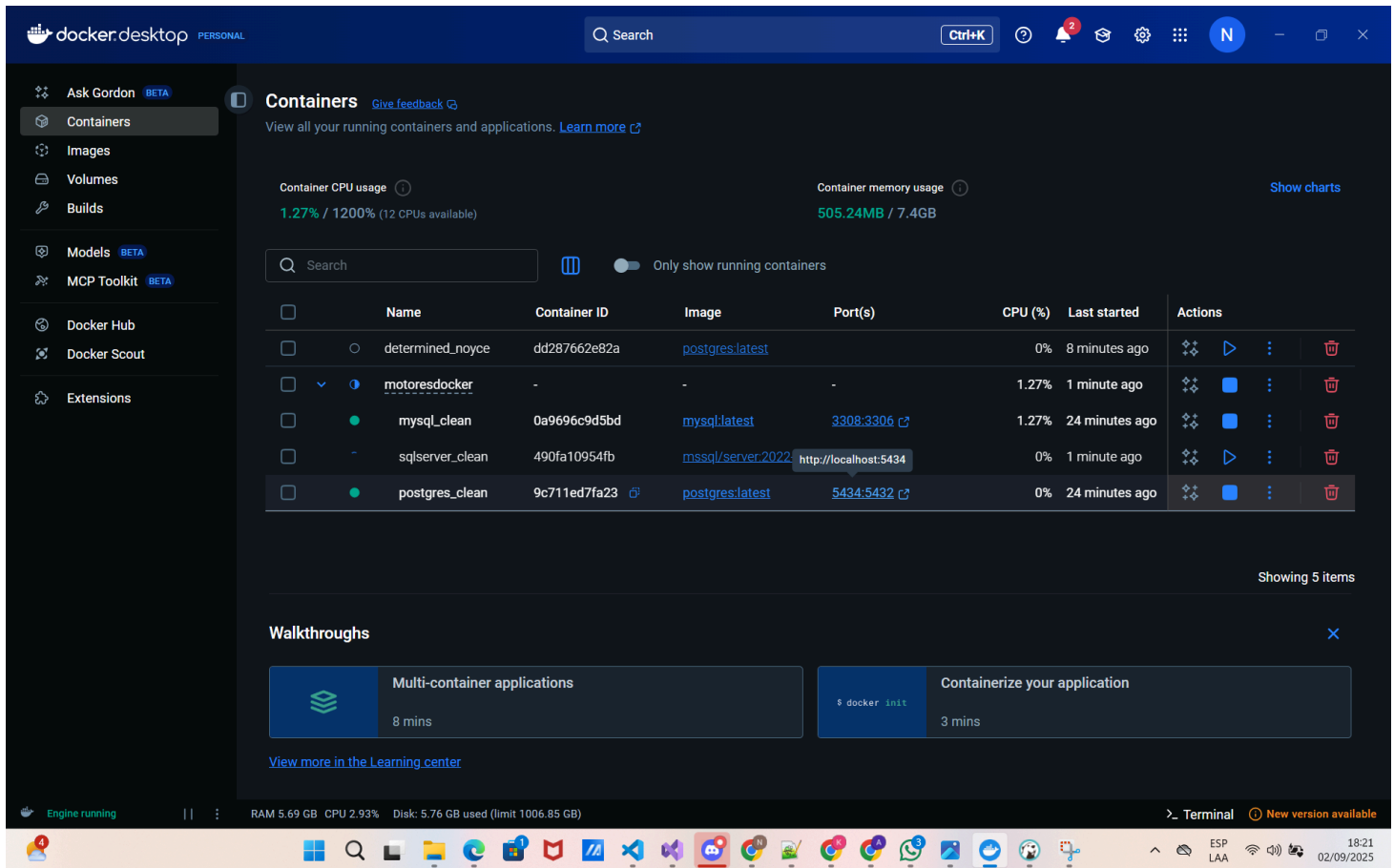
```
content-type: application/json; charset=utf-8
date: Tue, 09 Sep 2025 23:17:08 GMT
server: Microsoft-IIS/10.0
transfer-encoding: chunked
```
- Responses:** A table showing the response code (200) and description (OK).

The browser's taskbar at the bottom shows various application icons and the system clock indicating 18:18 on 09/09/2025.



1. Motores de Bases de Datos en Docker

Vista general de contenedores en Docker Desktop: En la captura se ve el panel Containers de Docker Desktop con los motores de base de datos ejecutándose en contenedores separados. Esta vista confirma que los contenedores están arriba y listos para aceptar conexiones externas desde herramientas cliente o desde aplicaciones.

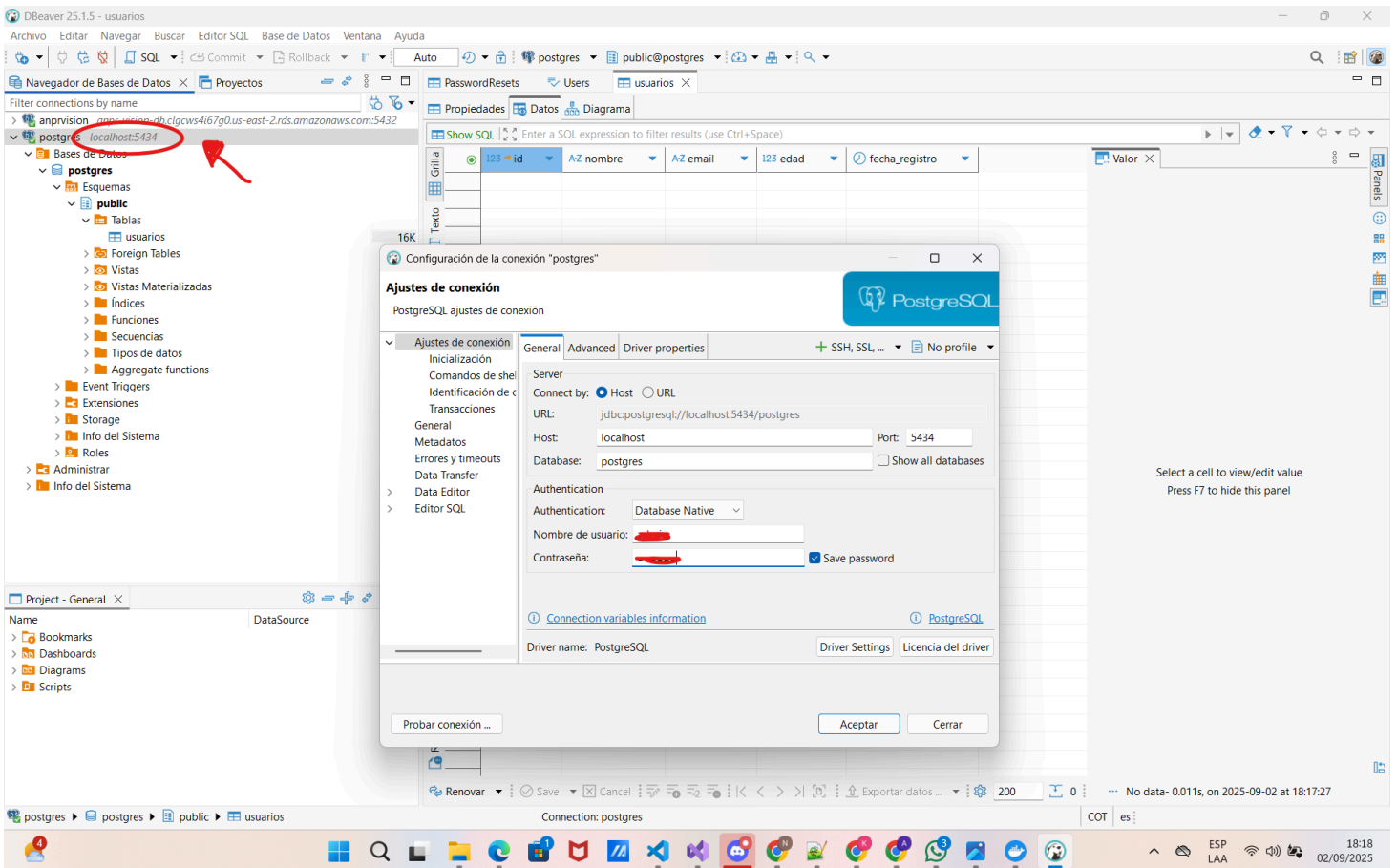
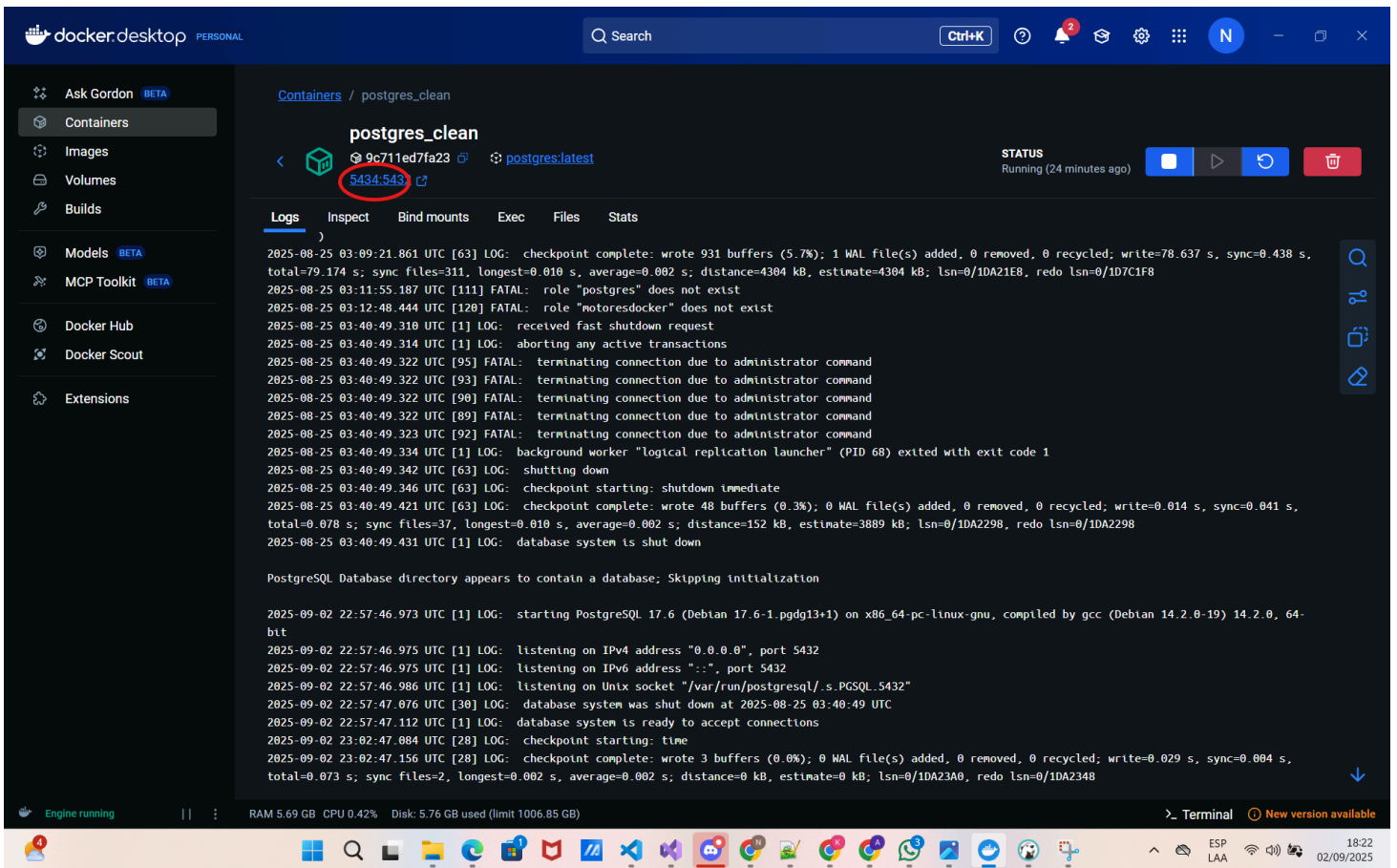


The screenshot shows the Docker Desktop interface. The left sidebar contains navigation options: Ask Gordon, Containers, Images, Volumes, Builds, Models, MCP Toolkit, Docker Hub, Docker Scout, and Extensions. The main panel is titled 'Containers' and shows a summary of container usage: CPU usage at 1.27% / 1200% (12 CPUs available) and memory usage at 505.24MB / 7.4GB. Below this is a table of running containers with columns for Name, Container ID, Image, Port(s), CPU (%), Last started, and Actions. The table lists five containers: 'determined_noyce' (postgres:latest), 'motoresdocker' (empty), 'mysql_clean' (mysql:latest), 'sqlserver_clean' (mssql/server:2022), and 'postgres_clean' (postgres:latest). The 'postgres_clean' container is highlighted. At the bottom, there are 'Walkthroughs' for 'Multi-container applications' and 'Containerize your application'. The status bar at the bottom indicates 'Engine running' and system resources: RAM 5.69 GB, CPU 2.93%, Disk 5.76 GB used (limit 1006.85 GB).

Name	Container ID	Image	Port(s)	CPU (%)	Last started	Actions
determined_noyce	dd287662e82a	postgres:latest		0%	8 minutes ago	[Icons]
motoresdocker	-	-	-	1.27%	1 minute ago	[Icons]
mysql_clean	0a9696c9d5bd	mysql:latest	3308:3306	1.27%	24 minutes ago	[Icons]
sqlserver_clean	490fa10954fb	mssql/server:2022	http://localhost:5434	0%	1 minute ago	[Icons]
postgres_clean	9c711ed7fa23	postgres:latest	5434:5432	0%	24 minutes ago	[Icons]

Conexión a PostgreSQL

La parte superior muestra los logs del contenedor de PostgreSQL indicando que el servidor se levantó correctamente y quedó a la espera de conexiones. Abajo se ve la conexión configurada en el cliente (DBeaver/pgAdmin) usando Host localhost, Puerto 5434, y las credenciales definidas en el contenedor. Esta combinación verifica que el mapeo de puertos es el correcto y que el motor en Docker responde como si fuera una instalación local, permitiendo explorar esquemas, tablas y ejecutar consultas sin salir del entorno de contenedores.



Conexión a MySQL

 **docker desktop** PERSONAL

Containers

Images

Volumes

Builds

Models BETA

MCP Toolkit BETA

Docker Hub

Docker Scout

Extensions

Ask Gordon BETA

Containers

Search

Ctrl+K



Containers / mysql_clean

mysql_clean

 **0a9696c9d5bd**  [mysql:latest](#)

STATUS

Running (36 minutes ago)



Logs

Inspect

Bind mounts

Exec

Files

Stats

2025-08-25T02:53:24.767793Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Bind-address: '::' port: 33060, socket: /var/run/mysqld/mysqld.sock

2025-08-25T02:53:24.769207Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '9.4.0' socket: '/var/run/mysqld/mysqld.sock' port: 3306 MySQL Community Server - GPL.

'/var/lib/mysql/mysql.sock' -> '/var/run/mysqld/mysqld.sock'

2025-08-25T03:40:38.784259Z 0 [System] [MY-013172] [Server] Received SHUTDOWN from user cvia user signal>. Shutting down mysqld (Version: 9.4.0).

2025-08-25T03:40:40.483444Z 0 [System] [MY-010910] [Server] /usr/sbin/mysqld: Shutdown complete (mysqld 9.4.0) MySQL Community Server - GPL.

2025-08-25T03:40:40.483513Z 0 [System] [MY-015816] [Server] MySQL Server - end.

2025-09-02 22:57:46+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 9.4.0-1.el9 started.

2025-09-02 22:57:47+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'

2025-09-02 22:57:47+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 9.4.0-1.el9 started.

'/var/lib/mysql/mysql.sock' -> '/var/run/mysqld/mysqld.sock'

2025-09-02T22:57:48.167182Z 0 [System] [MY-015815] [Server] MySQL Server - start.

2025-09-02T22:57:48.886623Z 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 9.4.0) starting as process 1

2025-09-02T22:57:48.886772Z 0 [System] [MY-015590] [Server] MySQL Server has access to 12 logical CPUs.

2025-09-02T22:57:48.886821Z 0 [System] [MY-015590] [Server] MySQL Server has access to 8131940352 bytes of physical memory.

2025-09-02T22:57:48.939466Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.

2025-09-02T22:57:50.023830Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.

2025-09-02T22:58:40.278887Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self signed.

2025-09-02T22:58:40.278888Z 0 [System] [MY-013602] [Server] channel mysql_main configured to support TLS. Encrypted connections are now supported for this channel.

2025-09-02T22:58:40.278895Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is accessible to all OS users. Consider choosing a different directory.

2025-09-02T22:58:40.278900Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Bind-address: '::' port: 33060, socket: /var/run/mysqld/mysqld.sock

2025-09-02T22:58:40.278901Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '9.4.0' socket: '/var/run/mysqld/mysqld.sock' port: 3306 MySQL Community Server - GPL.

2025-09-02T23:28:57.257321Z 11 [Warning] [MY-013360] [Server] Plugin sha256_password reported: 'sha256_password' is deprecated and will be removed in a future release. Please use caching_sha2_password instead'

2025-09-02T23:30:02.643381Z 12 [Warning] [MY-013360] [Server] Plugin sha256_password reported: 'sha256_password' is deprecated and will be removed in a future release. Please use caching_sha2_password instead'

2025-09-02T23:30:47.620402Z 13 [Warning] [MY-013360] [Server] Plugin sha256_password reported: 'sha256_password' is deprecated and will be removed in a future release. Please use caching_sha2_password instead'

Engine running

RAM 5.75 GB CPU 2.42% Disk: 5.76 GB used (limit 1006.85 GB)

Terminal

New version available



ESP LAA

18:34

02/09/2025

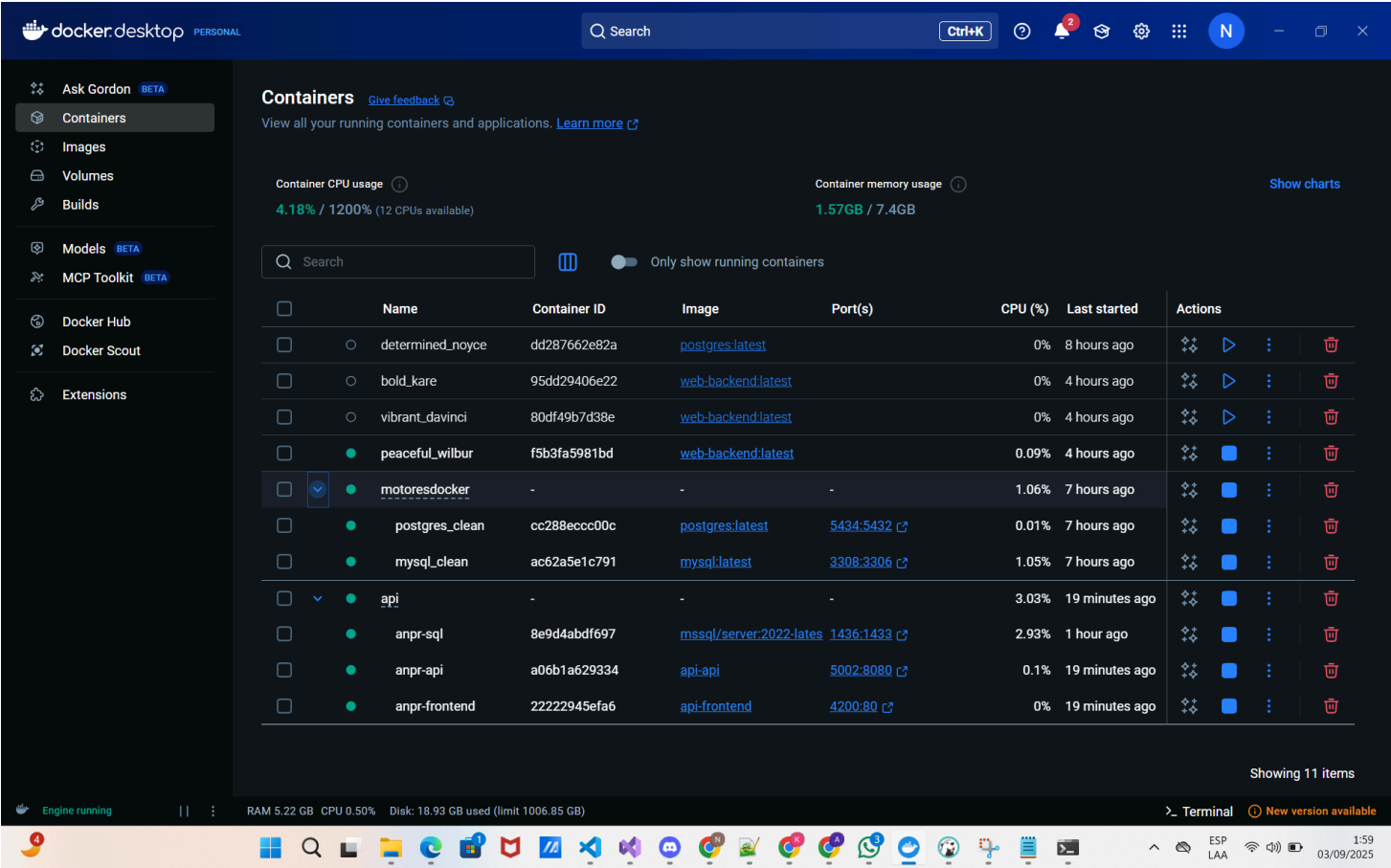
The screenshot shows the DBeaver 25.1.5 interface. The 'sys_config' table is selected in the 'Proyectos' pane. The 'Propiedades' pane shows the table's properties, including the 'sys_config' table name, InnoDB engine, and utf8mb4 charset. The 'Columns' pane displays the table's structure with 4 columns: 'variable' (varchar(128)), 'value' (varchar(128)), 'set_time' (timestamp), and 'set_by' (varchar(128)). The 'value' column has a default of 'CURRENT_TIMESTAMP' and a comment 'DEFAULT_GENERATED on update CURRENT_TIMESTAMP'. The 'set_by' column is highlighted in blue.

The screenshot displays the Docker Desktop interface with the 'Containers' tab selected. A container named 'sqlserver_clean' is running. The 'Logs' tab is active, showing a series of log entries. The container ID '490fa10954fb' is highlighted with a red circle. The logs indicate the initialization of SQL Server, including setting the instance name '490fa10954fb', clearing the tempdb database, and starting the availability replica manager. An error message is also present: 'ERROR: Unable to set system administrator password: Password validation failed. The password does not meet SQL Server password policy requirements because it is too short. The password must be at least 8 characters...'. The system tray at the bottom shows the time as 18:37 on 02/09/2025.

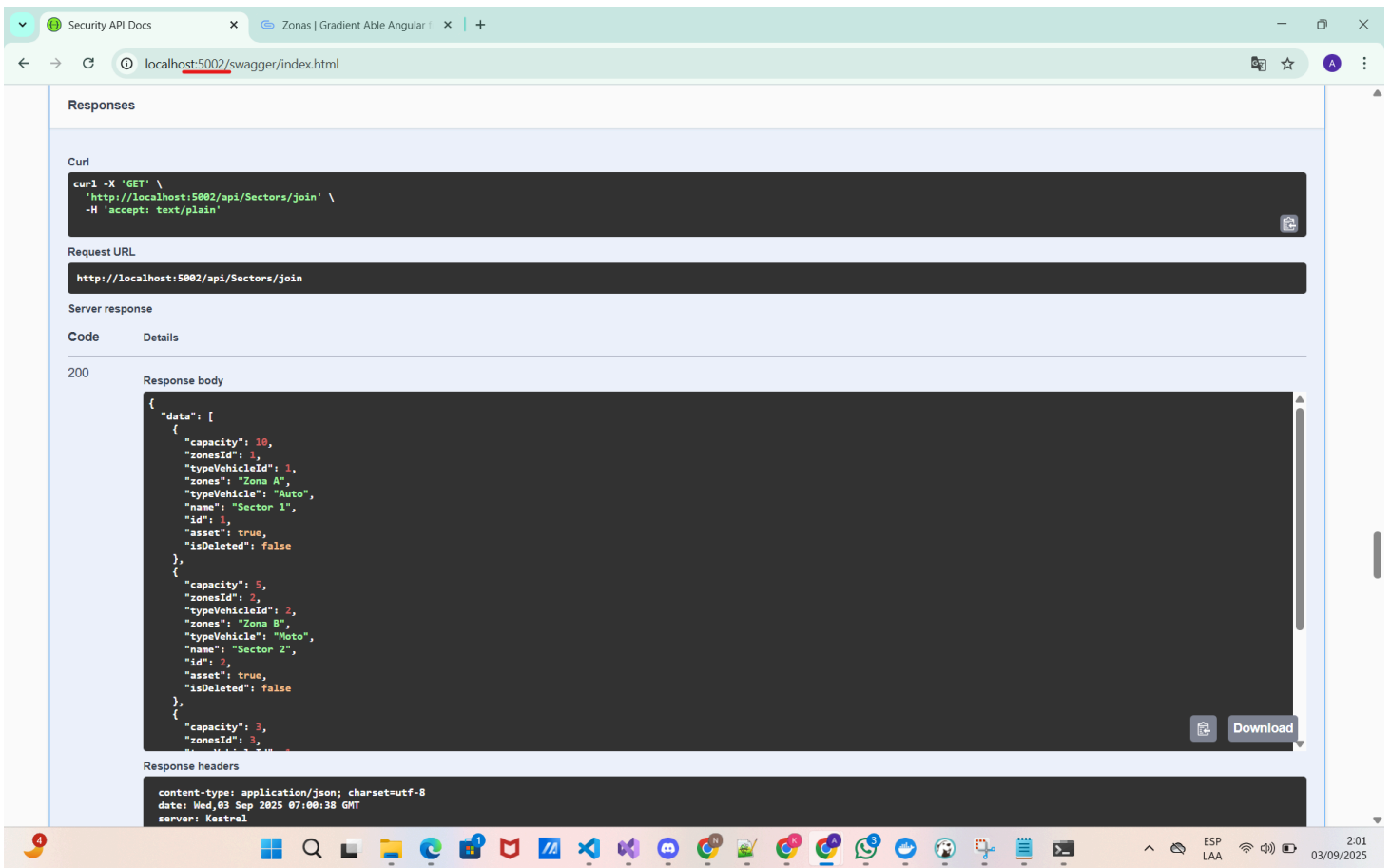
The screenshot shows the DBeaver 25.1.5 interface. The 'Navegador de Bases de Datos' (Database Navigator) pane on the left displays a tree view of the 'master' database. The 'master' database is selected, and its 'dbo' schema is also selected. The 'Properties' pane on the right shows the 'master' database details, including the 'Name' and 'Database description'. The 'Schemas' pane at the bottom shows the 'dbo' schema selected. The 'Project - General' pane at the bottom shows the 'DataSource' tab.

2. Backend y Frontend en Docker apuntando a la base de datos Docker

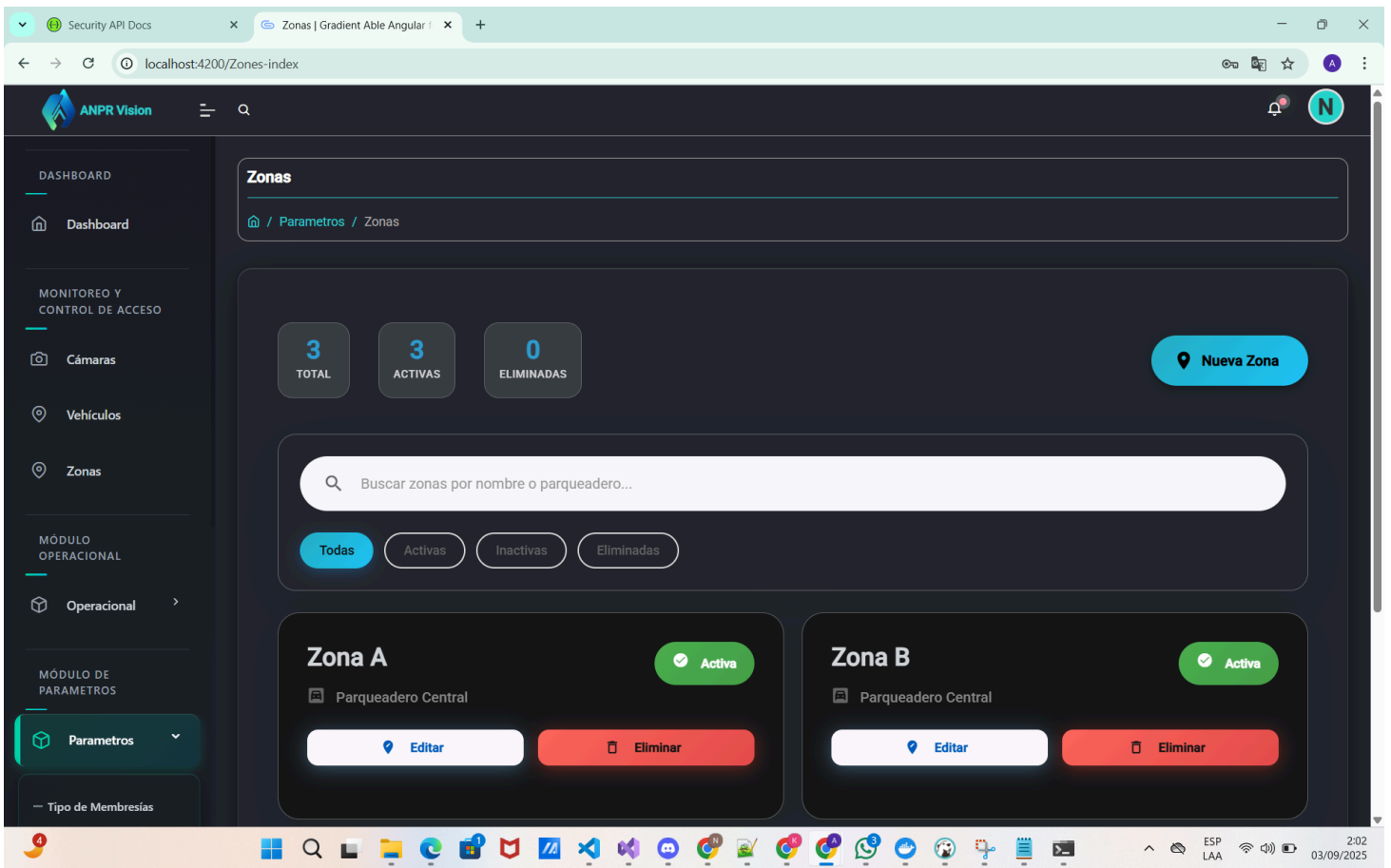
Vista general: contenedores de apps y motores de BD: La captura de Docker Desktop muestra ejecutándose, al mismo tiempo, los contenedores del backend, del frontend y de los motores de base de datos. Esta vista confirma que todos los servicios están arriba, comparten la misma red de Docker y pueden comunicarse por nombre de servicio.



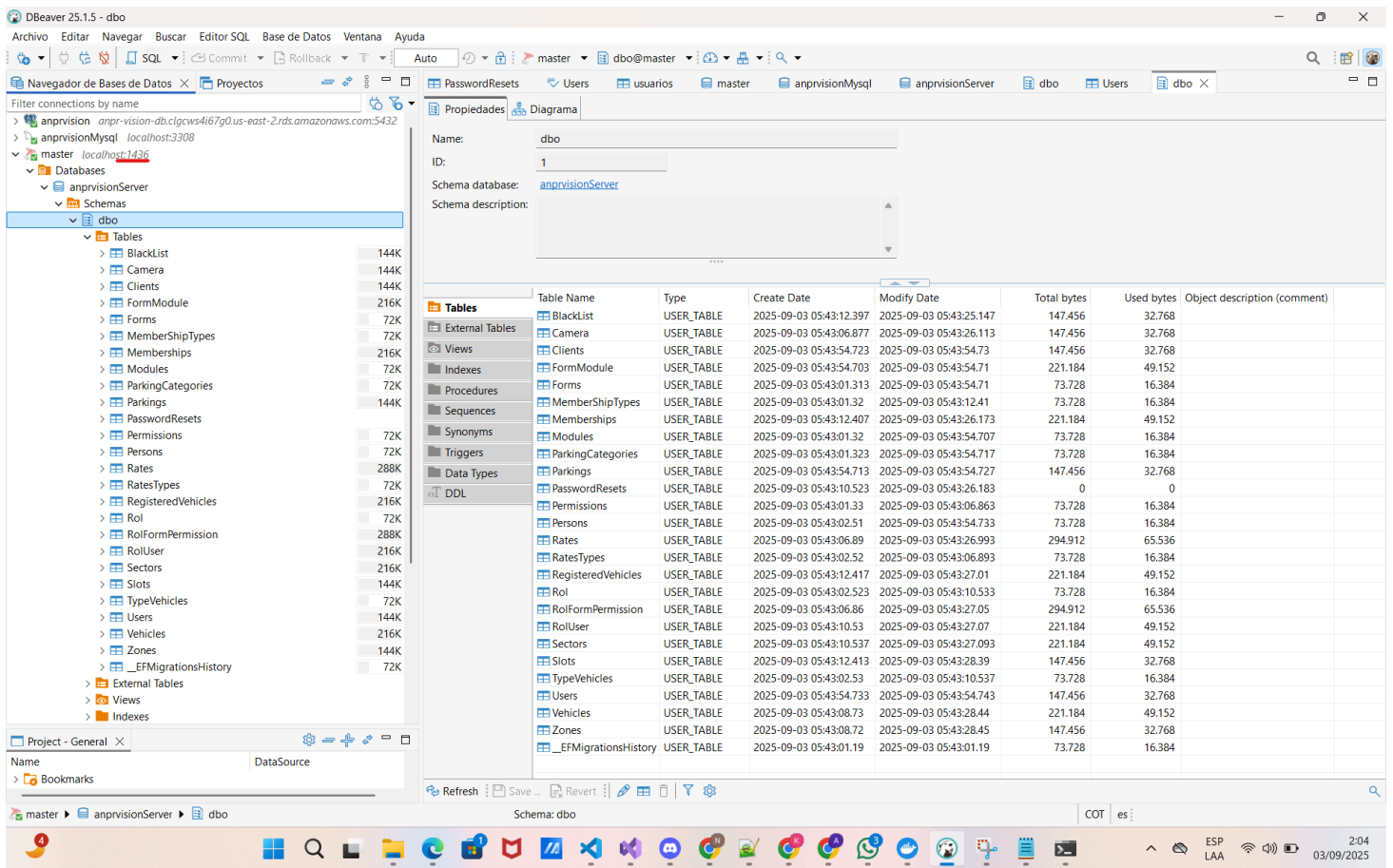
Prueba del backend con Swagger: En la sección inferior se abre Swagger UI de la API corriendo dentro del contenedor. Se ejecuta un endpoint (por ejemplo, /api/Sectors/join) y se observa el cURL, el Request URL y una respuesta 200 OK. Esto valida que el backend está correctamente expuesto hacia el host y que consulta la base de datos que también se ejecuta en Docker.



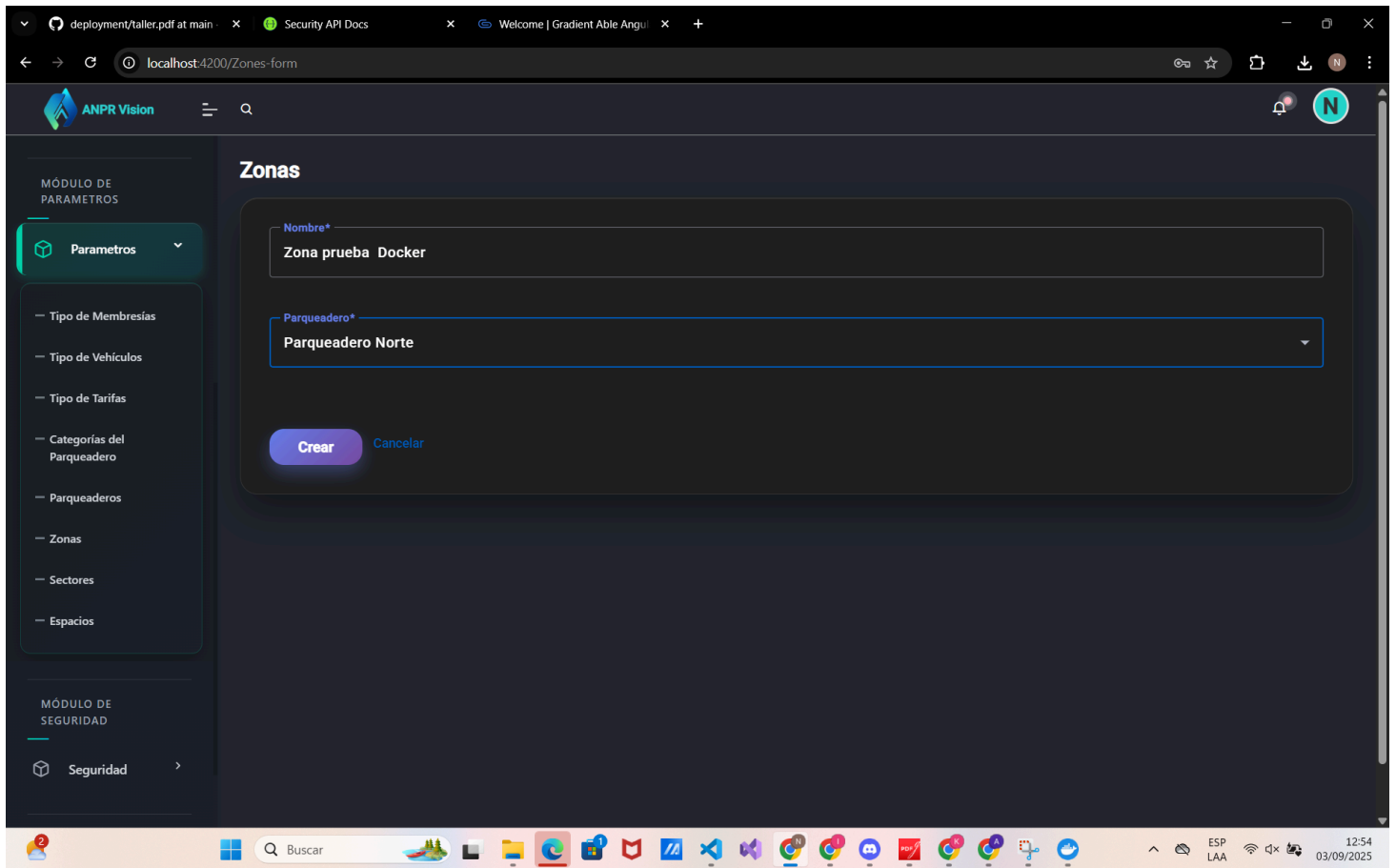
Frontend operativo y conectado al backend: La siguiente captura muestra el frontend en la pantalla de Zonas. Se visualizan tarjetas como Zona A y Zona B con sus estados, junto con filtros y acciones (Editar, Eliminar). Este resultado evidencia que el frontend está consumiendo la API y renderizando datos reales provenientes de la base de datos del entorno Docker.



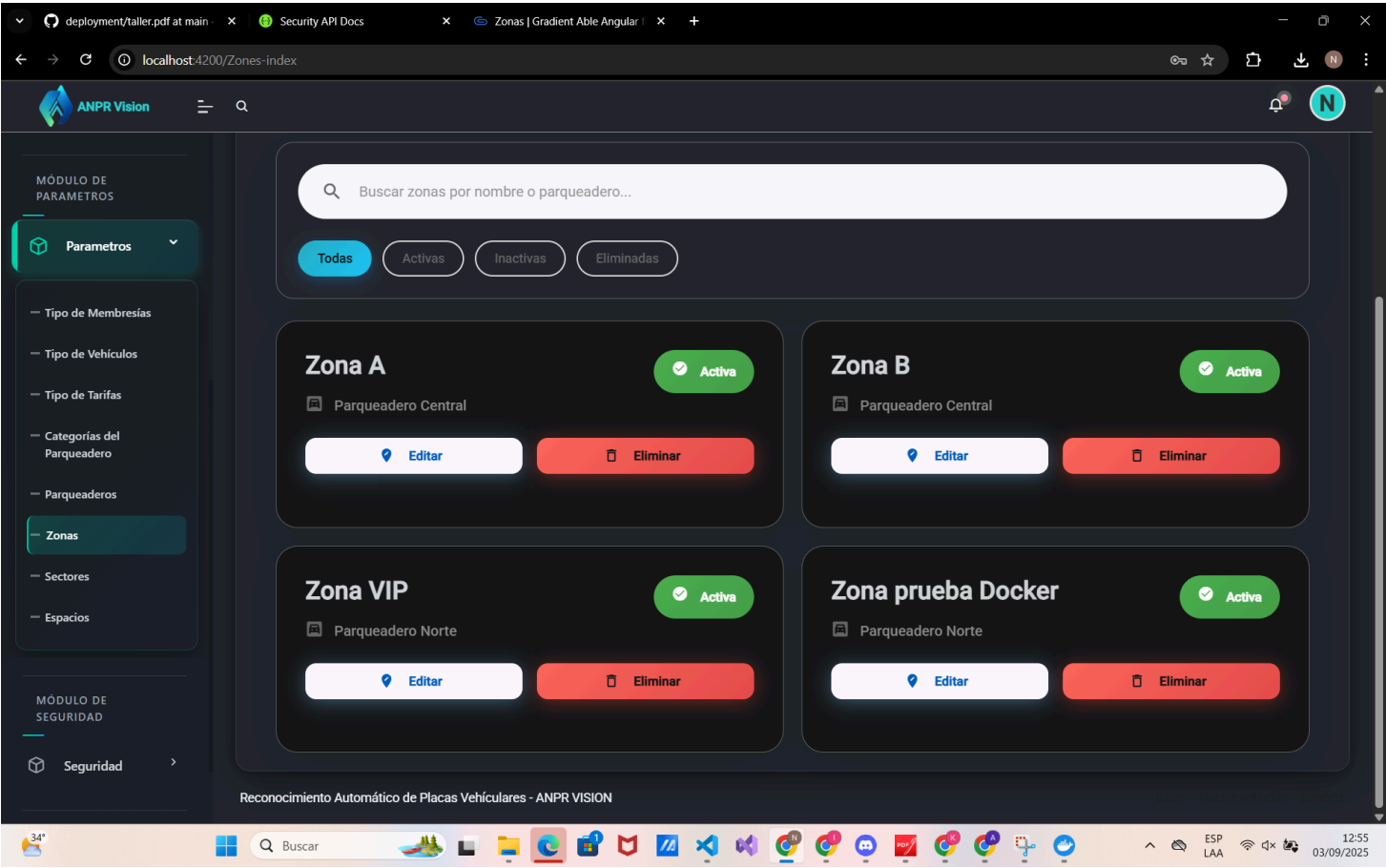
Verificación de datos en el cliente SQL: En la parte inferior se ve el cliente de base de datos (DBeaver) conectado al motor en Docker. Se inspeccionan esquemas y tablas del proyecto (por ejemplo, Zonas, Sectors, MembershipTypes, Rates, etc.). Esta comprobación cierra el circuito end-to-end: los datos mostrados en el frontend son los mismos que residen en la BD de Docker y que la API consulta en tiempo real.



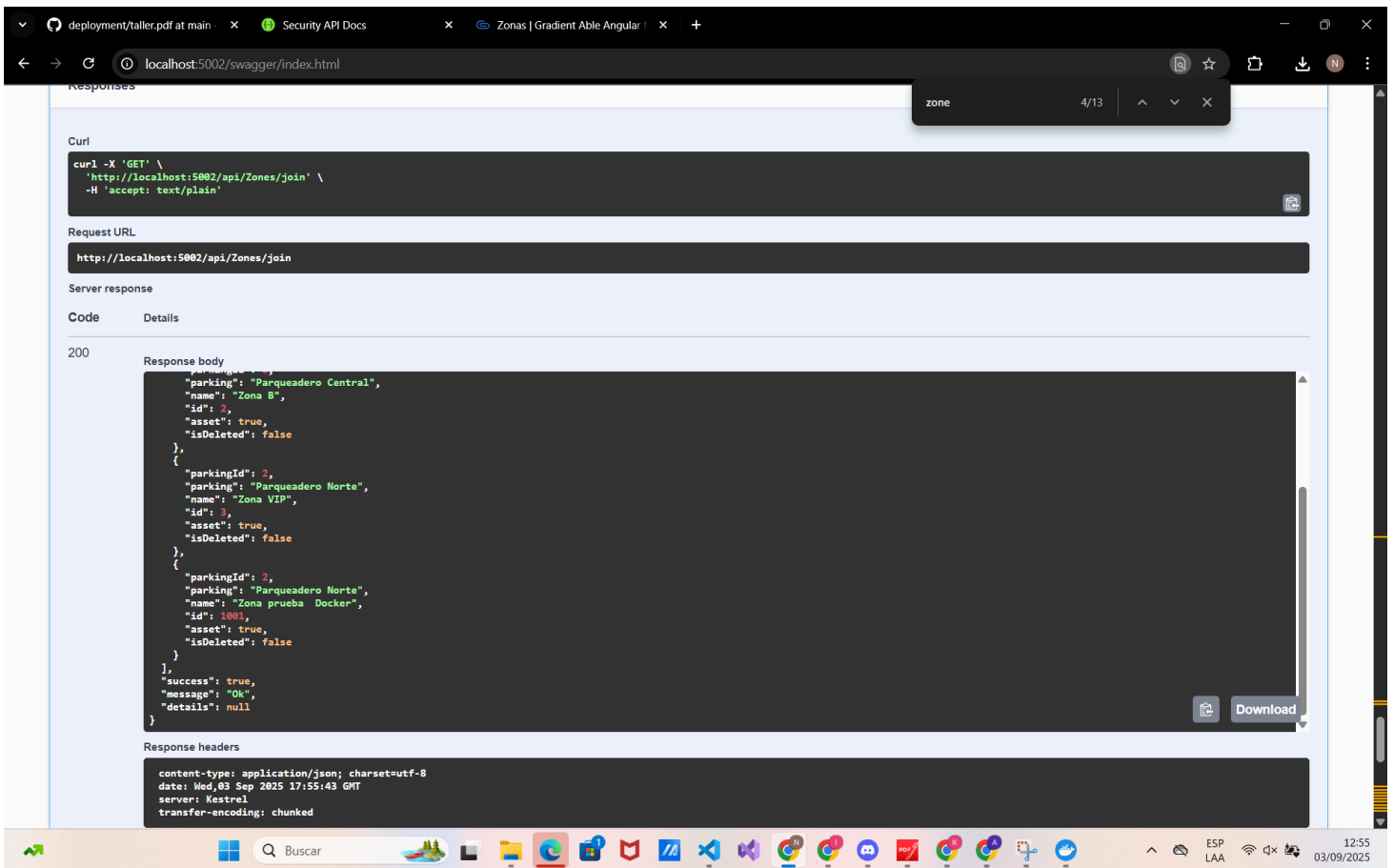
Realizamos un registro nuevo desde el fronted apuntando al backend montado en docker.



Registro creado



Verificamos que el registro se ha creado



Ubuntu

La imagen muestra la UI de Swagger en <https://localhost:8443/swagger/index.html> y, a la izquierda, el panel Security de DevTools indicando “This page is secure (valid HTTPS)”. Se ve que el certificado es válido y confiable, emitido por la CA de mkcert del equipo local. La sección Resources confirma que todos los recursos de la página se sirven de forma segura (sin contenido mixto).

Conclusión del proceso:

Se construyó un contenedor “todo en uno” (Ubuntu + SQL Server + [ASP.NET](#) Core + Nginx) con Kestrel escuchando en HTTP :5000 y Nginx haciendo la terminación TLS en :443 (expuesto como :8443 en el host).

Se resolvieron los problemas iniciales de arranque (elección del Web.dll y runtime .NET 9.0), se configuró la conexión a SQL Server interno (127.0.0.1,1433 con TrustServerCertificate/Encrypt=False), se añadió mkcert para un certificado confiable en local, se montó en Nginx y se ajustaron cabeceras de seguridad (HSTS y CSP: upgrade-insecure-requests).

El resultado final es una API servida por HTTPS válido en <https://localhost:8443>, con redirección limpia desde / a Swagger y sin advertencias de seguridad.

The screenshot shows a web browser window with the URL `localhost:8443/swagger/index.html`. The browser's DevTools console is open, displaying a security overview with a green message: "This page is secure (valid HTTPS)". The Swagger UI is visible, showing the API title "Security API v1 OAS 3.0" and a list of endpoints under the "BlackList" section. The endpoints include GET /api/BlackList/join, GET /api/BlackList/select, GET /api/BlackList/{id}, DELETE /api/BlackList/{id}, POST /api/BlackList, and PUT /api/BlackList. An "Authorize" button is visible in the top right corner of the Swagger UI.

Security overview

This page is secure (valid HTTPS).

Certificate - valid and trusted

The connection to this site is using a valid, trusted server certificate issued by mcert.LAPTOP-DDPC2U6E\Natalia osorio@LAPTOP-DDPC2U6E (natalia osorio).

View certificate

Connection - secure connection settings

The connection to this site is encrypted and authenticated using TLS 1.3, X25519, and AES_128_GCM.

Resources - all served securely

All resources on this page are served securely.

Security API v1 OAS 3.0

/swagger/v1/swagger.json

API para gestión de usuarios, roles y permisos

Tu Nombre - Website

Send email to Tu Nombre

Authorize

BlackList

GET /api/BlackList/join

GET /api/BlackList/select

GET /api/BlackList/{id}

DELETE /api/BlackList/{id}

POST /api/BlackList

PUT /api/BlackList

available